Project Report For University Library Management System

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1. Introduction

This report presents a relational database designed for a University Library Management System. The system was implemented using Python and SQLite and includes several related tables such as Students, Authors, Books, Library_Branches, Book_Copies, and Loans.

The purpose of the database is to manage library operations including storing student information, managing books and authors, tracking library branches, and recording book borrowing transactions. To preserve data integrity, the system uses primary keys, foreign keys, constraints, and a composite key.

All data used in this project were synthetically generated using Python libraries to simulate realistic library operations without using real personal information.

2. Data Generation Process

The dataset utilised in this study was created using Python. Fake student data, including names, email addresses, nationalities, and dates of birth, was generated using the Faker library. Additional information, including book genres, publication years, page numbers, book conditions, loan and return dates, and fine amounts, was generated using the random module.

The database contains approximately:

- **1000+ Students**
- **200 Authors**
- **500 Books**
- **3 Library Branches**
- **1500 Book Copies**
- **2000 Loan Records**

Some records include NULL values in return_date to represent books that have not yet been returned, simulating realistic library borrowing situations.

3. Database Schema Design

The database is designed to reduce redundancy and maintain data integrity. Information is distributed across multiple related tables and connected using foreign keys.

Students (1) → Loans (N)
Authors (1) → Books (N)
Books (1) → Book_Copies (N)
Library_Branches (1) → Book_Copies (N)
Book_Copies (1) → Loans (N)

3.1 Students Table Schema

Attributes:
• student_id – INTEGER PRIMARY KEY
• first_name – TEXT
• last_name – TEXT
• gender – TEXT CHECK('Male','Female','Other')
• year_of_study – INTEGER CHECK BETWEEN 1 AND 4
• date_of_birth – DATE
• email – TEXT UNIQUE
• membership_type – TEXT CHECK('Undergraduate','Postgraduate','PhD')

3.2 Authors Table Schema

Attributes:
• author_id – INTEGER PRIMARY KEY
• first_name – TEXT
• last_name – TEXT
• nationality – TEXT

3.3 Books Table Schema

Attributes:
• book_id – INTEGER PRIMARY KEY
• title – TEXT
• genre – TEXT CHECK('Science','Engineering','Law','Arts','Medicine','Other')
• publication_year – INTEGER
• isbn – TEXT UNIQUE
• pages – INTEGER
• author_id – INTEGER REFERENCES Authors(author_id)

3.4 Library_Branches Table Schema

Attributes:
• branch_id – INTEGER PRIMARY KEY
• branch_name – TEXT
• city – TEXT
• capacity – INTEGER

3.5 Book_Copies Table Schema

Attributes:

• copy_id – INTEGER PRIMARY KEY
• book_id – INTEGER REFERENCES Books(book_id)
• branch_id – INTEGER REFERENCES Library_Branches(branch_id)
• condition – TEXT CHECK('New','Good','Fair','Poor')
• available – BOOLEAN

3.6 Loans Table Schema

Attributes:

• student_id – INTEGER REFERENCES Students(student_id)
• copy_id – INTEGER REFERENCES Book_Copies(copy_id)
• loan_date – DATE
• due_date – DATE
• return_date – DATE
• fine_amount – REAL

Composite Primary Key:
(student_id, copy_id, loan_date)

4. Justification for Table Design

In order to reduce redundancy and enhance scalability, the database adheres to normalisation principles. Repeatedly storing author information is avoided by separating authors from books. Multiple physical copies of the same book can exist in various branches thanks to Book_Copies. Financial penalties and borrowing history may be tracked in detail thanks to the Loans table.

5. Ethical and Data Privacy Considerations

Python modules like Faker and random were utilised to create all of the artificial data used in this project. No actual personal data was gathered or kept in the database.

Synthetic data guarantees ethical data use for scholarly objectives and privacy protection. Additional security mechanisms like encryption, secure authentication, role-based access control, and adherence to data protection laws like GDPR would also be necessary in real-world applications.

6. Conclusion

This project shows a thorough understanding of structured data production, foreign key relationships, relational database design, SQL constraints, and composite primary keys. The solution accurately depicts a realistic university library administration database and satisfies all assignment requirements.

7. GitHub Link

https://github.com/samiaarif199510/SQL-Project